

TLS (Transport Layer Security)

What Is TLS?

TLS is a protocol that:

Encrypts communication between client and server.

Without TLS:

- Anyone on network can read data.
- Passwords, cookies, tokens are visible.

With TLS:

- Data is encrypted.
- Only client and server can read it.

TLS is what makes **HTTPS** secure.

Where TLS Fits in Network Stack

Full flow when opening a website:

1. ARP → Resolve gateway MAC
2. DNS → Get server IP
3. TCP → Establish connection
4. TLS → Secure the connection
5. HTTP → Send actual data

TLS sits **on top of TCP**.

It cannot work without TCP.

Why TLS Is Needed

Problems TLS solves:

1. Confidentiality
→ Encrypts data
2. Integrity
→ Detects tampering
3. Authentication
→ Verifies server identity

Without TLS:

Anyone in between can spy or modify traffic.

What Happens During TLS Handshake

TLS handshake happens before encrypted data starts.

Let's break TLS 1.2 first (easier to understand).

TLS 1.2 Handshake Steps

1. Client Hello

Client sends:

- Supported TLS versions
- Supported cipher suites
- Random number (Client Random)
- Extensions (including SNI)

SNI = domain name (very important)

Client is saying:

“Here's what I support.”

2. Server Hello

Server replies with:

- Selected TLS version
- Selected cipher suite
- Server Random

Server is saying:

“I choose this encryption method.”

Now both sides have:

- Client Random
- Server Random

3. Certificate

Server sends its certificate.

Certificate contains:

- Public key
- Domain name
- Issuer (CA)
- Expiry date

Browser checks:

- Is certificate trusted?
- Does domain match?
- Is it expired?

If valid → continue

If not → show warning

4. Server Key Exchange

Server sends key exchange parameters (usually ECDHE).
This is used to securely generate shared secret.

5. Client Key Exchange

Client sends its part of key exchange.

Now both sides calculate:

Shared Secret

From this shared secret + random numbers → session keys are derived.

6. Change Cipher Spec

Both sides switch to encrypted mode.

From now on:

Everything is encrypted.

7. Encrypted Application Dataa

Now actual HTTP data starts.

Wireshark shows:

“Application Data”

But you cannot read content.

TLS 1.3 (Modern Version)

TLS 1.3 is faster and more secure.

Differences:

- Fewer steps
- No separate Server Key Exchange
- Handshake messages encrypted faster
- Better forward secrecy

TLS 1.3 flow:

Client Hello
Server Hello
Encrypted Extensions
Certificate
Certificate Verify
Finished
Encrypted Application Data

Much faster than TLS 1.2.

What Are Cipher Suites?

Cipher suite defines:

- Key exchange algorithm (ECDHE, RSA, etc.)
- Encryption algorithm (AES, ChaCha20)
- Hash algorithm (SHA256, etc.)

Example:

TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

Meaning:

- ECDHE → Key exchange
- RSA → Authentication
- AES-128-GCM → Encryption
- SHA256 → Hash

What Is SNI?

SNI = Server Name Indication.

It sends domain name inside Client Hello.

Why needed?

Because: Many websites share same IP.

Without SNI: Server wouldn't know which certificate to send.

That's why HTTPS fails when connecting via raw IP sometimes.

Why Ping Works but HTTPS Needs Domain

Ping:

Uses ICMP.

No certificate.

No TLS.

HTTPS:

Needs correct certificate.

Certificate must match domain.

That's why direct IP often fails.

What TLS Protects (And What It Doesn't)

Encrypted:

- URL path
- Headers
- Cookies
- Body data
- Passwords

NOT encrypted:

- Destination IP
- TCP port
- TLS handshake (partially visible)
- SNI (visible in TLS 1.2)

That's why network analysts can still see:

- Which domain you're visiting
- But not what page content is

What Wireshark Shows

Filter: tls

You will see:

Client Hello
Server Hello
Certificate
Key Exchange
Change Cipher Spec
Application Data

But you will NOT see:

Plain HTTP content (because encrypted).

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Capturing from wlan0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

11s

No.	Time	Source	Destination	Protocol	Length	Info
118	18.698526663	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	595	Client Hello (SNI=collector.bsg.brave.com)
123	18.177198923	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
128	18.284151549	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	550	Certificate, Server Key Exchange, Server Hello Done
131	18.211432798	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
132	18.212111776	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	185	Application Data
133	18.212333328	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	435	Application Data
134	18.212468603	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TCP	1374	49088 -> 443 [ACK] Seq=2308 Ack=4269 Win=70656 Len=1288 TSval=2797964599 TSecr=2511910880 [TCP PDU reassembled in 136]
138	18.213109456	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	856	Application Data
143	18.428018966	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
149	18.436018542	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	559	Certificate, Server Key Exchange, Server Hello Done
151	18.438467833	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

Frame 132: Packet, 185 bytes on wire (1480 bits), 185 bytes captured (1480 bits) on interface wlan0, id 0

Ethernet II, Src: AzureWaveTec.co:11:2b (14:11:33:c9:11:2b), Dst: 2a:89:57:c4:75:df (2a:89:57:c4:75:df)

Internet Protocol Version 6, Src: 2409:40d6:119e:bdc:e11a:f988:d1f5:4bc8, Dst: 64:ff9b::22d2:2c78

Transmission Control Protocol, Src Port: 49088, Dst Port: 443, Seq: 1866, Ack: 4269, Len: 99

Transport Layer Security

[Stream index: 4]

- TLSv1.2 Record Layer: Application Data Protocol: HyperText Transfer Protocol 2

Content Type: Application Data (23)

Version: TLS 1.2 (0x0303)

Length: 94

Encrypted Application Data: 0000000000000001cd7e2cebc565337381394591bf1ad0a30097397a4d8f2793a5cd6c67c605cf52230cb67feec24cdbf81debc

[Application Data Protocol: HyperText Transfer Protocol 2]

Application Data Protocol (tls.app_data_proto)

Packets: 3479 - Displayed: 225 (6.5%)

Profile: Default

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Capturing from wlan0

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11s

No.	Time	Source	Destination	Protocol	Length	Info
118	18.698526663	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	595	Client Hello (SNI=collector.bsg.brave.com)
123	18.177198923	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
128	18.284151549	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	550	Certificate, Server Key Exchange, Server Hello Done
131	18.211432798	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
132	18.212111776	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	185	Application Data
133	18.212333328	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	435	Application Data
134	18.212468603	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TCP	1374	49088 -> 443 [ACK] Seq=2308 Ack=4269 Win=70656 Len=1288 TSval=2797964599 TSecr=2511910880 [TCP PDU reassembled in 136]
138	18.213109456	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	856	Application Data
143	18.428018966	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
149	18.436018542	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	559	Certificate, Server Key Exchange, Server Hello Done
151	18.438467833	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

Frame 131: Packet, 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits) on interface wlan0, id 0

Ethernet II, Src: AzureWaveTec.co:11:2b (14:11:33:c9:11:2b), Dst: 2a:89:57:c4:75:df (2a:89:57:c4:75:df)

Internet Protocol Version 6, Src: 2409:40d6:119e:bdc:e11a:f988:d1f5:4bc8, Dst: 64:ff9b::22d2:2c78

Transmission Control Protocol, Src Port: 49088, Dst Port: 443, Seq: 1734, Ack: 4269, Len: 126

Transport Layer Security

[Stream index: 4]

- TLSv1.2 Record Layer: Handshake Protocol: Client Key Exchange

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 76

- Handshake Protocol: Client Key Exchange

Handshake Type: Client Key Exchange (16)

Length: 66

- EC Diffie-Hellman Client Params

- TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec

Content Type: Change Cipher Spec (20)

Version: TLS 1.2 (0x0303)

Length: 1

Change Cipher Spec Message

- TLSv1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 48

Handshake Protocol: Encrypted Handshake Message

wlan0<live capture in progress>

Packets: 3450 - Displayed: 217 (6.3%)

Profile: Default

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Capturing from wlan0

No.	Time	Source	Destination	Protocol	Length	Info
118	18.69852663	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	595	Client Hello (SNI=collector.bsg.brave.com)
123	18.177108923	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
131	18.211432796	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
132	18.21211776	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	185	Application Data
133	18.212333328	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	435	Application Data
134	18.212468603	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TCP	1374	49088 - 443 [ACK] Seq=2308 Ack=4269 Win=70656 Len=1288 TSval=2797964599 TSecr=2511910880 [TCP PDU reassembled in 136]
138	18.213109456	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	856	Application Data
143	18.428018966	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
149	18.436018542	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	559	Certificate, Server Key Exchange, Server Hello Done
151	18.438467833	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

Frame 128: Packet, 556 bytes on wire (4480 bits), 556 bytes captured (4480 bits) on interface wlan0, id 0
 Ethernet II, Src: 2a:89:57:c4:75:df (2a:89:57:c4:75:df), Dst: AzureWaveTec-c9:11:2b (14:13:33:c9:11:2b)
 Internet Protocol Version 6, Src: 64:ff9b::22d2:2c78, Dst: 2409:40d6:119e:bdc:e11a:f98d:d1f5:4bc8
 Transmission Control Protocol, Src Port: 443, Dst Port: 49088, Seq: 3805, Ack: 1734, Len: 464
 [4 Reassembled TCP Segments (3808 bytes): #123(1155), #125(1268), #127(1268), #128(1171)]
 Transport Layer Security
 [Stream index: 4]
 - TLSv1.2 Record Layer: Handshake Protocol: Certificate
 Content Type: Handshake (22)
 Version: TLS 1.2 (0x0303)
 Length: 3893
 - Handshake Protocol: Certificate
 Handshake Type: Certificate (11)
 Length: 3799
 Certificates Length: 3796
 Certificates (3796 bytes)
 - Transport Layer Security
 [Stream index: 4]
 - TLSv1.2 Record Layer: Handshake Protocol: Server Key Exchange
 Content Type: Handshake (22)
 Version: TLS 1.2 (0x0303)
 Length: 333
 - Handshake Protocol: Server Key Exchange
 Handshake Type: Server Key Exchange (12)
 Length: 329
 EC Diffie-Hellman Server Params
 - TLSv1.2 Record Layer: Handshake Protocol: Server Hello Done
 Content Type: Handshake (22)
 Version: TLS 1.2 (0x0303)
 Length: 4
 - Handshake Protocol: Server Hello Done
 Handshake Type: Server Hello Done (14)
 Length: 0

Packet (550 bytes) Reassembled TCP (3808 bytes)

Packets: 3435 - Displayed: 212 (6.2%) Profile: Default

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Capturing from wlan0

No.	Time	Source	Destination	Protocol	Length	Info
118	18.6985266	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	595	Client Hello (SNI=collector.bsg.brave.com)
123	18.1771089	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
128	18.2041151	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	556	Certificate, Server Key Exchange, Server Hello Done
131	18.2111127	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
132	18.2121117	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	185	Application Data
133	18.2123333	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	435	Application Data
134	18.2124686	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TCP	1374	49088 - 443 [ACK] Seq=2308 Ack=4269 Win=70656 Len=1288 TSval=2797964599 TSecr=2511910880 [TCP PDU reassembled in 136]
136	18.2131094	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	856	Application Data
143	18.4280189	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	1354	Server Hello
149	18.4360185	64:ff9b::22d2:2c78	2409:40d6:119e:bdc...	TLSv1..	559	Certificate, Server Key Exchange, Server Hello Done
151	18.4384678	2409:40d6:119e:bdc...	64:ff9b::22d2:2c78	TLSv1..	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

Frame 123: Packet, 1354 bytes on wire (10832 bits), 1354 bytes captured (10832 bits) on interface wlan0, id 0
 Ethernet II, Src: 2a:89:57:c4:75:df (2a:89:57:c4:75:df), Dst: AzureWaveTec-c9:11:2b (14:13:33:c9:11:2b)
 Internet Protocol Version 6, Src: 64:ff9b::22d2:2c78, Dst: 2409:40d6:119e:bdc:e11a:f98d:d1f5:4bc8
 Transmission Control Protocol, Src Port: 443, Dst Port: 49088, Seq: 1, Ack: 1734, Len: 1288
 Transport Layer Security
 [Stream index: 4]
 - TLSv1.2 Record Layer: Handshake Protocol: Server Hello
 Content Type: Handshake (22)
 Version: TLS 1.2 (0x0303)
 Length: 108
 - Handshake Protocol: Server Hello
 Handshake Type: Server Hello (2)
 Length: 104
 Version: TLS 1.2 (0x0303)
 Random: c54eaf570601ae3b53859ba63da7a1cfc6b40e42aa7f889b457063ad9823d9
 Session ID Length: 32
 Session ID: f09c7dbc8c3ecb33ec8a75f70dc73178462282f14bc1a6efb1f0dda6690155
 Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)
 Compression Method: null (0)
 Extensions Length: 32
 Extension: server_name (len=0)
 Extension: ec_point_formats (len=2)
 Extension: renegotiation_info (len=1)
 Extension: application_layer_protocol_negotiation (len=5)
 Extension: session_ticket (len=0)
 Extension: extended_master_secret (len=0)
 [JA3S FullString: 771,49199,0-11-65281-16-35-23]
 [JA3S: bfc9d56141386ee83b56cda231ccfc]
 TLS segment data (1155 bytes)

Packet (550 bytes) Reassembled TCP (3808 bytes)

Packets: 3273 - Displayed: 169 (5.2%) Profile: Default

Capturing from wlan0

No. Time Source Destination Protocol Length Info

118 18.8985266 2489:4d6e:119e:: 64:f9fb::22d2:2 TLSv1_ 595 Client Hello (SNI=collector.bsg.brave.com)

123 18.171989 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 1354 Server Hello

128 18.204151 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 550 Certificate, Server Key Exchange, Server Hello Done

131 18.2114327 2489:4d6e:119e:: 64:f9fb::22d2:2 TLSv1_ 212 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

132 18.2121117 2489:4d6e:119e:: 64:f9fb::22d2:2 TLSv1_ 185 Application Data

133 18.2123333 2489:4d6e:119e:: 64:f9fb::22d2:2 TLSv1_ 435 Application Data

134 18.2124666 2489:4d6e:119e:: 64:f9fb::22d2:2 TCP 1374 49088 -> 443 [ACK] Seq=2308 Ack=4269 Win=7056 Len=1288 TSval=2797964599 TSecr=251191880 [TCP PDU reassembled]

136 18.2131994 2489:4d6e:119e:: 64:f9fb::22d2:2 TLSv1_ 855 Application Data

143 18.4280189 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 1354 Server Hello

149 18.4366185 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 550 Certificate, Server Key Exchange, Server Hello Done

151 18.4384678 2489:4d6e:119e:: 64:f9fb::22d2:2 TLSv1_ 212 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

153 18.5112890 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 290 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message

154 18.5112892 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 155 Application Data

155 18.512985 64:f9fb::22d2:2 2489:4d6e:119e:: TLSv1_ 124 Application Data

Frame 118: Packet, 595 bytes on wire (4760 bits), 595 bytes captured (4760 bits) on interface wlan0, id 0
Ethernet II, Src: AzureWaveTec_c8:11:2b (14:13:3c:c9:11:2b), Dst: 2a:89:57:c4:75:d7 (2a:89:57:c4:75:d7)
Internet Protocol Version 6, Src: 2489:4d6e:119e:bdc:e11a:P988:d1f5:4bc8, Dst: 64:f9fb::22d2:2
Transmission Control Protocol, Src Port: 49992, Dst Port: 443, Seq: 1289, Ack: 1, Len: 509
[2 Reassembled TCP Segments (1797 bytes) : #117(1289), #118(509)]

Transport Layer Security
(Stream index: 3)
--TLSv1_2 Record Layer: Handshake Protocol: Client Hello
Content Type: Handshake (22)
Version: TLS 1.0 (0x0301)
Length: 1792
Handshake Protocol: Client Hello
--Handshake Type: Client Hello (1)
Length: 1788
Version: TLS 1.2 (0x0303)
Random: 604c483ccce96643ba35b4db1a93b5174bc4cf5762586396ac5428a2df
Session ID Length: 32
Session ID: f1cd675bf73d80342cef955d217bec6e99abc8987afd972903cab46be3d7198
Cipher Suites Length: 32
Cipher Suites (16 suites)
Compression Methods Length: 1
Compression Methods (1 method)
Extensions Length: 1683
Extension: Reserved (GREASE) (len=0)
Extension: signed_certificate_timestamp (len=0)
Extension: status_request (len=5)
Extension: session_ticket (len=0)
Random values used for deriving keys (its handshake.random.32.bytes)

Forward Secrecy (Important Security Concept)

Modern TLS uses ECDHE.

That means:

Even if someone steals server private key later,

old traffic cannot be decrypted.

That's called:

Forward Secrecy.

Very important in security.

Real-World TLS Use Cases

TLS is used in:

- HTTPS
- Secure email (SMTPS)
- VPN
- API communication
- Banking apps
- Mobile apps

Almost everything modern uses TLS.

TLS Summary in One Flow

Client: "Hello, I support these encryptions."

Server: "Okay, use this one. Here's my certificate."

Client: “Certificate verified. Let’s generate secret.”

Both: “Switch to encrypted mode.”

Now: Secure communication begins.

Big Picture

ARP → Find MAC

DNS → Find IP

TCP → Build connection

TLS → Secure connection

HTTP → Transfer data

TLS is the security layer of the modern internet.